ALAMEIN
INTERNATIONAL UNIVERSITY
جامعـة العلميـن الدوليـة

Phy 211

Lectures Notes

By

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Lecture 3

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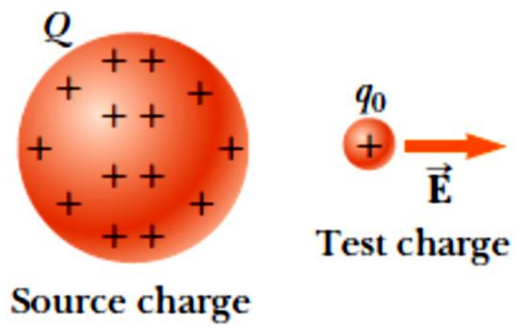
➢ The electric field $(\vec{E})$:

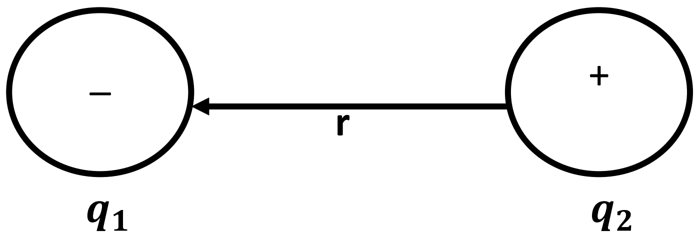
- An electric field is said to exist in the region of space around a charged object.
- The electric field exerts an electric force on any other charged object within the field.
- The magnitude of the electric field is the magnitude of the electric force on a positive unit test charge "q_o", divided by this charge

$$\boxed{E = \frac{F}{q_o}\ (\mathrm{N/C})}$$

$$\boxed{E = \frac{KQq_o/_{r^2}}{q_o} = \frac{KQ}{r^2}\ \ (\mathrm{N/C})}$$

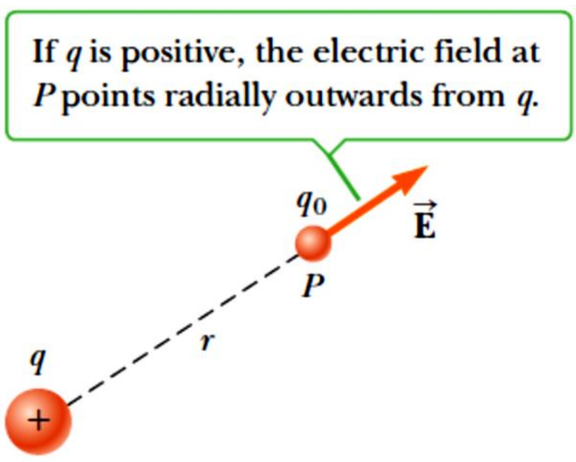


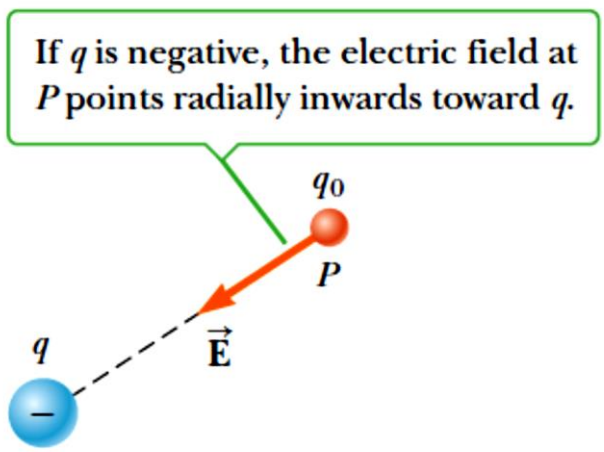
- The magnitude of electric field intensity at "q_2" due to the charge "q_1" is given by:



$$E = \frac{F}{|q_2|} = K\frac{|q_1||q_2|}{r^2|q_2|} = K\frac{|q_1|}{r^2}\ \ (\mathrm{N/C})$$

- The electric field is directed away from +ve charge. The electric field is directed towards –ve charge.





- The electric field at a point due to several charges is the vector sum of all the fields due to the several charges at this point.

$$\overrightarrow{\boldsymbol{E}_{net}} = \overrightarrow{\boldsymbol{E}_1} + \overrightarrow{\boldsymbol{E}_2} + \overrightarrow{\boldsymbol{E}_3} + \cdots$$

➢ Motion of charged particle inside an electric field:

- When a charged particle is placed in an electric field, it experiences an electrical force.
- The net force will cause the particle to accelerate according to Newton's Second Law.
- If the particle has a positive charge, its acceleration is in the direction of the field.
- If the particle has a negative charge, its acceleration is in the direction opposite the electric field.

$$\boxed{\boldsymbol{F} = \boldsymbol{E}|\boldsymbol{q}| = \boldsymbol{ma}}$$

F: force on the particle

q: charge of the particle

m: mass of the particle

a: particle acceleration $\left(\boldsymbol{a} = \frac{\Delta \boldsymbol{v}}{\Delta \boldsymbol{t}}\ rate\ of\ change\ of\ velocity\right)$

$$a = \frac{Eq}{m} \longrightarrow a \propto \frac{1}{m}$$

$$\because m_e \ll m_p$$

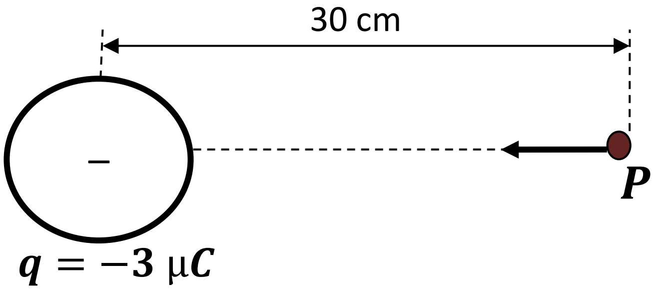
- If a proton and an electron is placed in the same field

$$\therefore a_e \gg a_p$$

This idea is used in charge accelerators.

- **Example 7:**

Find the magnitude and direction of the electric field $\vec{E}$ due to a charge q of magnitude -3 μC at a point P, 30 cm from the charge.



Solution

$$\left|\vec{E}\right| = K_e \frac{|q|}{r^2} = 9 \times 10^9 \frac{3 \times 10^{-6}}{0.3^2} = 3 \times 10^5\ N/C$$

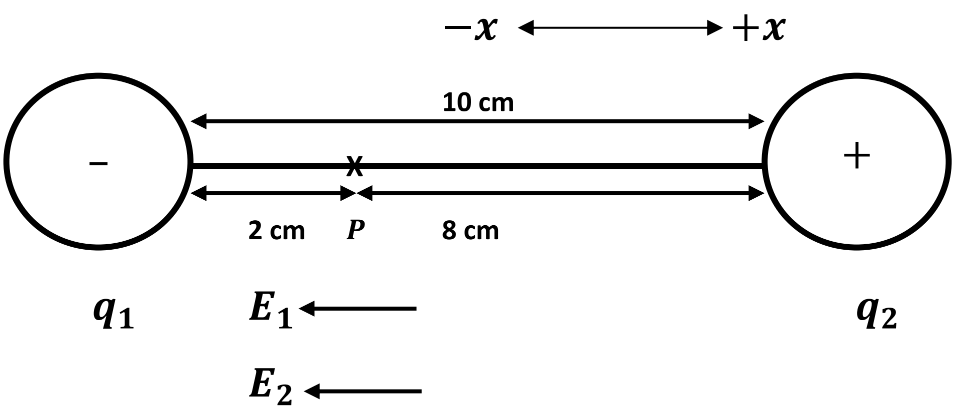
The electric field id directed from P towards the charge q.

- **Example 8:**

Two charges q_1 $(-25\ \mu\text{C})$ and q_2 $(50\ \mu\text{C})$ are separated by a distance of 10 cm,

a) What is the magnitude & the direction of electric field at a point "P" between the two charges at 2 cm to the right of q_1.
b) If an electron is placed at "P", what will be its acceleration (direction & magnitude) (mass of e = 9.1×10^{-31} kg & charge of e = 1.6×10^{-19} C)

Solution



a)

$$E_1 = K\frac{|q_1|}{0.02^2} = 9\times10^9\frac{25\times10^{-6}}{0.02^2} = 5.625\times10^8\ \mathrm{N/C}\quad (-x\ \ direction)$$

$$E_2 = K\frac{|q_2|}{0.08^2} = 9\times10^9\frac{50\times10^{-6}}{0.08^2} = 0.703\times10^8\ \mathrm{N/C}\ \ (-\ x\ \ direction)$$

$$E_p = -E_1 - E_2 = -5.625\times10^8 - 0.703\times10^8 = -6.328\times10^8\ \mathrm{N/C}$$

The field magnitude is $6.328\times10^8\ \mathrm{N/C}$ & is directed into the $-x$ direction.

b)

$$F_e = E_p|q_e| = m_e a_e$$

$$\therefore\ a_e = \frac{E_p|q_e|}{m_e} = \frac{(6.328\times10^8)\times(1.6\times10^{-19})}{(9.1\times10^{-31})} = 1.11\times10^{20}\ \mathrm{m/sec^2}$$

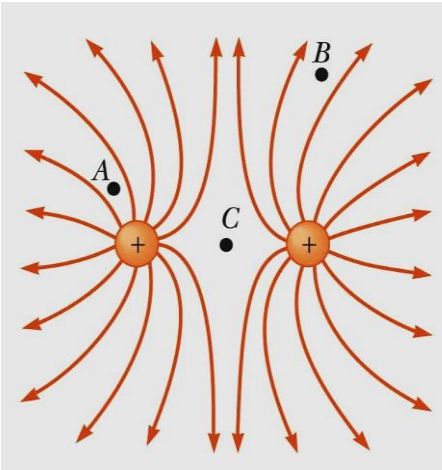
Since the electron is a $-ve$ charge it will move in the opposite direction of the field. So, it will be accelerated in the $+x$ direction.

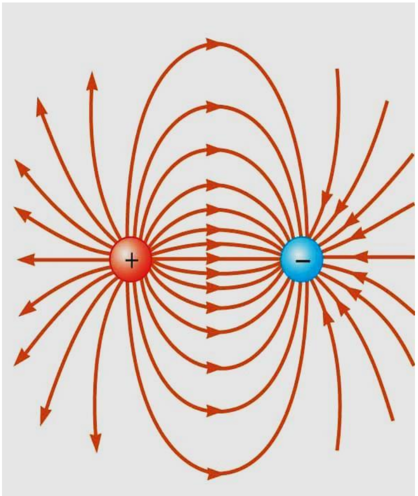
➢ Electric Field Lines:

- The electric field can be represented by **field lines** such that:

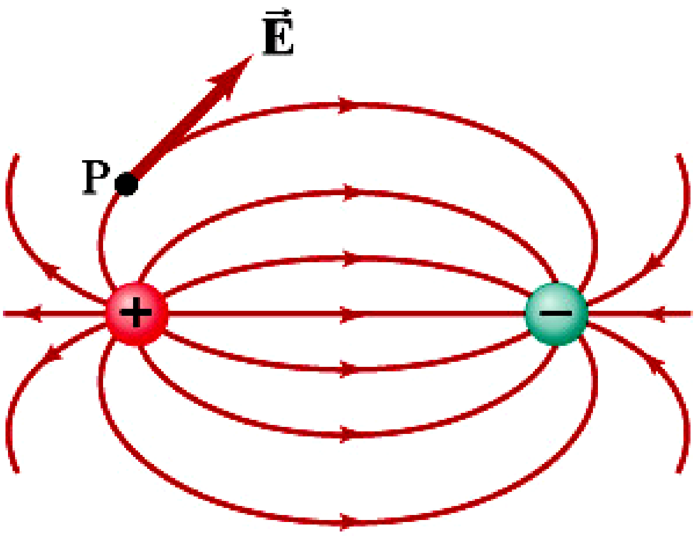
1- The lines must begin on a positive charge and terminate on a negative charge.

2- Field lines cannot intersect (even the charges are too close).

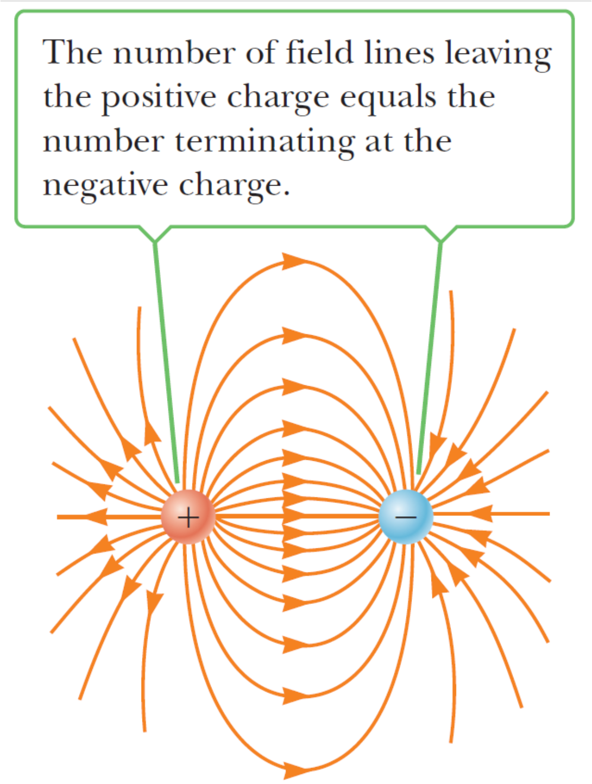


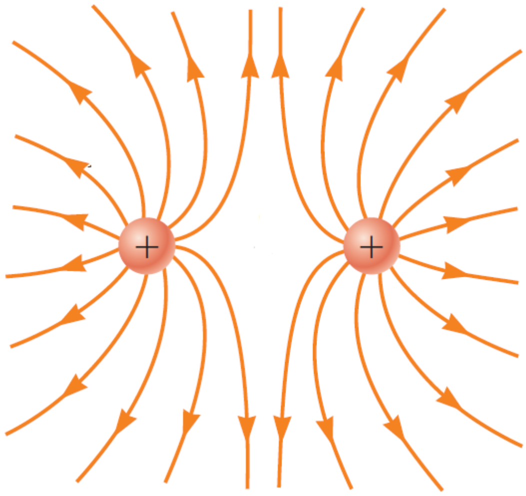


3- Direction of the electric field at any point is the is the tangent to the field line at this point.

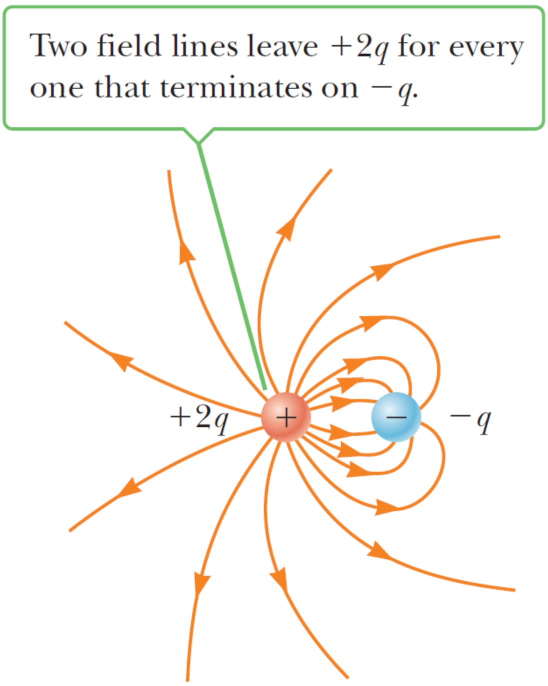


4- The number of lines leaving a positive charge or entering a negative charge is proportional to the magnitude of the charge.

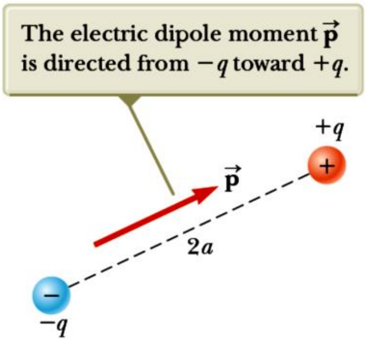




The electric field lines for two positive point charges.



➢ **Electric dipole:** It consists of two charge of equal magnitude and opposite sign separated by a distance $2a$. (cannot be separated)

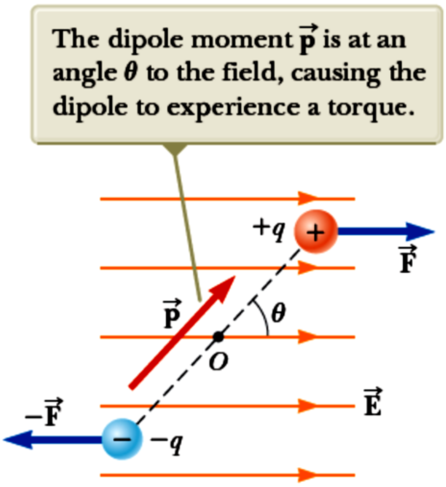


- **Electric Dipole moment:** it is the vector $\vec{p}$ directed from $-q$ toward $+q$ along the line joining the charges and having magnitude:

$$\boxed{\mathbf{p} = q \cdot 2a}$$

- When an electric dipole is placed in an electric field, the two charges experience two equal and opposite forces, such that the whole dipole is affected by a torque $\vec{\tau}$.

- As a result of this torque; the dipole tends to be aligned with the electric field direction. This phenomenon is called Polarization.



$$Torque \ = \ force \ \times normal\ distance\ between\ the\ two\ forces$$

$$\boxed{\begin{gathered}\tau \ = F \times (2a \sin\theta) = (qE) \times (2a \sin\theta) \ = \ 2aqE\sin\theta \ = \ \mathbf{p}E\sin\theta \\ \vec{\tau} = \vec{\mathbf{p}} \times \vec{E} \quad (N \cdot m)\end{gathered}}$$

Due to this tourque, the dipole acquires a potential energy given by:

$$\boxed{Pot\ Energy = \vec{\mathbf{p}} \cdot \vec{E} = \mathbf{p}E\cos\theta \quad (Joule)}$$